

# The Ephemeris

December 2014

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## Historic First Landing on a Comet November 12, 2014



*This image of comet 67P/Churyumov-Gerasimenko was taken by the Philae lander of the European Space Agency's Rosetta mission during Philae's descent toward the comet on Nov. 12, 2014. Philae's ROLIS camera took the image from a distance of approximately two miles (three kilometers) from the surface.  
Image Credit: ESA/Rosetta/Philae/DLR*

After more than a decade traveling through space, a robotic lander built by the European Space Agency has made the first-ever soft landing of a spacecraft on a comet. Mission controllers at ESA's mission operations center in Darmstadt, Germany, received a signal confirming that the Philae lander had touched down on comet 67P/Churyumov-Gerasimenko on Wednesday, Nov. 12, just after 8 a.m. PST/11 a.m. EST. Philae's mother ship, the Rosetta spacecraft, will remain in orbit around the comet through 2015. The orbiter will continue detailed studies as the comet approaches the sun and then moves away.

In addition to their well-deserved reputation as beautiful cosmic objects, comets hold vital clues about our solar system's history. They are considered primitive building blocks of the solar system that are literally frozen in time. Comets may have played a part in "seeding" Earth with water and, possibly, the basic ingredients for life.

Credit: nasa.gov.

### December 2014 Events

#### Saturday, Dec 06

Board Meeting: 6 -7:30pm

General Mtg: 7:30-9:30pm

#### Sunday, Dec 07

Solar observing: 2-4pm

Fix-It Day: 2-4pm

#### Friday, Dec 12

Beginner Astronomy Class: 7pm

In-Town Star party (Houge): 7-9pm

#### Saturday, Dec 13

Starry Nights Star Party (RCDO): 7-9pm

#### Saturday, Dec 20

Henry Coe—Dark Sky Weekend

#### Saturday, Jan 03

Board Meeting: 6-7:30pm

General Mtg: 7:30-9:30pm

#### Sunday, Jan 04

Solar observing: 2-4pm

Fix-It Day: 2-4pm

#### Friday, Jan 09

Beginner Astronomy Class: 7pm

In-Town Star party (Houge): 7-9pm

#### Saturday, Jan 17

Henry Coe—Dark Sky Weekend

*SJAA events are subject to cancellation due to weather. Please visit website for up-to-date info.*

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## Philae Lander Sniffed Out Organics in Comet's Atmosphere

The first probe ever to land on the surface of a comet performed some serious science before going into hibernation. Europe's Philae lander found organic molecules in the comet's atmosphere and discovered that the frigid object's surface is as hard as ice.

On Nov. 12, the European Space Agency's Philae became the first probe to softly land on the face of a comet. After being released from the Rosetta orbiter, the lander actually bounced off Comet 67P/Churyumov-Gerasimenko twice before coming to its current less-than-ideal resting spot. Because of the low sunlight conditions, Philae went into hibernation after only about 57 hours on the comet when its primary batteries depleted. But the probe still beamed back a wealth of science during its short initial life on the icy body.

While it will take scientists a while to sift through the data collected by Philae, it looks like the probe has sent home some interesting new results. Before shutdown, one of Philae's instruments managed to "sniff" the first organic molecules detected in the atmosphere of the comet, officials with the DLR German Aerospace Center said. However, scientists still aren't sure what kind of organics — carbon-containing molecules that are the building blocks of life on Earth — were found.

Philae also found that Comet 67P/C-G's surface is harder than researchers initially thought it would be. Before the probe's battery ran out, Rosetta mission controllers commanded Philae to hammer into the

surface of the comet, and found that the cosmic body is probably as hard as ice, according to ESA.

"Although the power of the hammer was gradually increased, we were not able to go deep into the surface," research team leader Tilman Spohn, of the DLR Institute of Planetary Research said in a statement. "We have acquired a wealth of data, which we must now analyze."

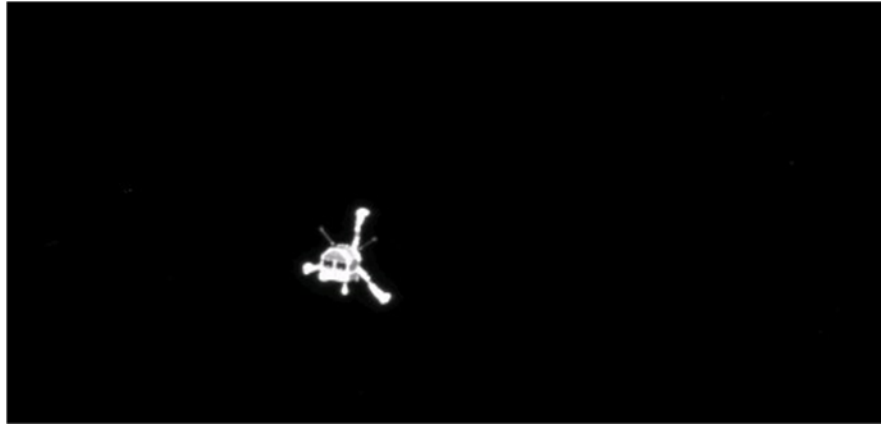
An instrument onboard Rosetta has also recently made an important discovery for scientists interested in the composition of the comet's ice and its potential implications for how Earth became a watery world.

ESA officials are still hopeful that Philae could get in touch again as Comet 67P/C-G makes its way around the sun. It's possible that the sunlight and

temperature conditions around the lander could change as the comet gets closer to our star, allowing Philae to potentially come back to life. "I'm very confident that Philae will resume contact with us and that we will be able to operate the instruments again," DLR lander project manager Stephan Ulamec said in a statement.

Rosetta is expected to stay with the comet as it makes its closest approach to the sun in August 2015. The orbiter is scheduled to study the spacecraft through at least December 2015.

Credit ESA



Philae - Photo Credit ESA



Philae touchdown on the comet: Credit ESA

## Jupiter's Red Spot is Likely a Sunburn, Not a Blush



*Image credit: NASA/JPL-Caltech/ Space Science Institute*

The ruddy color of Jupiter's Great Red Spot is likely a product of simple chemicals being broken apart by sunlight in the planet's upper atmosphere, according to a new analysis of data from NASA's Cassini mission. The results contradict the other leading theory for the origin of the spot's striking color -- that the reddish chemicals come from beneath Jupiter's clouds.

Baines and JPL colleagues Bob Carlson and Tom Momary arrived at their conclusions using a combination of data from Cassini's December 2000 Jupiter flyby and laboratory experiments.

In the lab, the researchers blasted ammonia and acetylene gases -- chemicals known to exist on Jupiter -- with ultraviolet light, to simulate the sun's effects on these materials at the extreme heights of clouds in the Great Red Spot. This produced a reddish material, which the team compared to the Great Red Spot as observed by Cassini's Visible and Infrared Mapping Spectrometer (VIMS). They found that the light-scattering properties of their red concoction nicely matched a model of the Great Red Spot in which the red-colored material is confined to the uppermost reaches of the giant cyclone-like feature.

"Our models suggest most of the Great Red Spot is actually pretty bland in color, beneath the upper cloud layer of reddish material," said Baines. "Under the reddish 'sunburn' the clouds are probably whitish or grayish." A coloring agent confined to the top of the clouds would be inconsistent with the competing theory, which posits that the spot's red color is due to upwelling chemicals formed deep beneath the visible cloud layers, he said. If red material were being transported from below, it should be present at other altitudes as well, which would make the red spot redder

still. Jupiter is composed almost entirely of hydrogen and helium, with just a sprinkling of other elements. Scientists are interested in understanding what combinations of elements are responsible for the hues seen in Jupiter's clouds, as this would provide insights into the giant planet's make-up.

Baines and colleagues initially set out to determine if the Great Red Spot's color might derive from sun-induced breakdown of a more complex molecule, ammonium hydrosulfide, which makes up one of Jupiter's main cloud layers. They quickly found that instead of a red color, the products their experiment produced were a brilliant shade of green. This surprising negative result prompted the researchers to try simple combinations of ammonia with hydrocarbons that are common at Jupiter's high altitudes. Breaking down ammonia and acetylene with ultraviolet light turned out to best fit the data collected by Cassini.

The Great Red Spot is a long-lived feature in Jupiter's atmosphere that is as wide as two earths. Jupiter possesses three main cloud layers, which occupy specific altitudes in its skies; from highest to lowest they are: ammonia, ammonium hydrosulfide and water clouds.

As for why the intense red color is seen only in the Great Red Spot and a few much smaller spots on the planet, the researchers think altitude plays a key role. "The Great Red Spot is extremely tall," Baines said. "It reaches much higher altitudes than clouds elsewhere on Jupiter."

The team thinks the spot's great heights both enable and enhance the reddening. Its winds transport ammonia ice particles higher into the atmosphere than usual, where they are exposed to much more of the sun's ultraviolet light. In addition, the vortex nature of the spot confines particles, preventing them from escaping. This causes the redness of the spot's cloud tops to increase beyond what might otherwise be expected.

Other areas of Jupiter display a mixed palette of oranges, browns and even shades of red. Baines says these are places where high, bright clouds are known to be much thinner, allowing views to depths in the atmosphere where more colorful substances exist.

*Credit: NASA/JPL-Caltech/ Space Science Institute*



# SJAA Solar Volunteers at Scouting Encampment at Alameda Fairground

From Bill O'Neil



On October 11, 2014, the San Francisco Bay Area Boy & Girl Scouts held an Encampment at the Alameda County Fairgrounds in Pleasanton. It was a STEM event (Science, Technology, Engineering & Math) and the Science topic this year was Astronomy. Nate McKenna, the Science Lead, stopped by our September SJAA meeting and put out a call for volunteers to help out with Solar Telescopes.



*Girl Scouts viewing through SJAA scope*



*Wolfe & Roland Math Class 3*

Telescopes to view sunspots came from Eastbay Astronomical Society and Tri-Valley Star Gazers Astronomy Club. Bill O'Neil and Wolf Witt represented SJAA with two Lunt Hydrogen-alpha telescopes to see solar flares and prominences, which were a big hit with the kids! We brought the NASA "Our Magnetic Sun" poster used at SJAA Solar Sundays, and gave talks to the scouts throughout the day on the dynamic Sun, the spectrum of light, how a H-alpha telescope works, and even how to calculate the time it takes sunlight to get to Earth.

Over 1,500 scouts attended and 500 signed up for the Astronomy track. Since we were in a large open area, we also attracted many other scouts who stopped by to talk to us and look through the telescopes. It was a big event with exhibits and talks by Lawrence Livermore Labs, Align Technology (3-D), and NASA's Brian Day (SJAA member) and former astronaut, Tammy Jernigan, who spoke at lunch about her five Space Shuttle missions.



## Solar Sundays

From Michael Packer

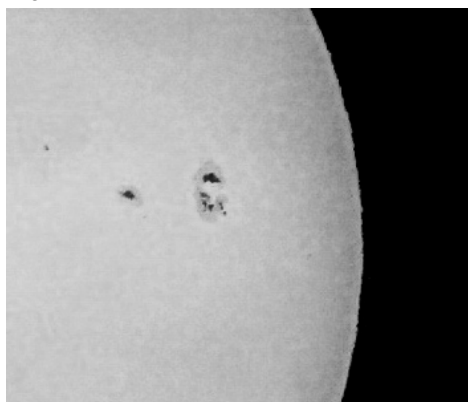
Solar Sunday at Houge saw about 15 folks stop by – Join us next month December 7th (Dec 14th is back up date in case of low sunspot count or weather).



*Solar Sunday at Houge; Regulars Bill, Kevin, Jack (relaxing), Carl*

Last month [October] the eclipse and sunspot AR-2192 were superb – unlikely but perhaps the sunspot will last a month, long enough to spin back in view. If it does look for it on ~ Nov 18th.

*Editor's Comment: As noted by Michael Packer in the SJAA Blog, sunspot AR2192 did return, re-numbered to AR2209 [photo below] for its second trip around the sun. The AR2209 photo was taken at the Santa Cruz lighthouse solar observing session on Nov 16th.*



*Photo: Sun Spot AR2209  
Photo credit: Bill Seiler*



*Photo: Bill Seiler, Michael Packer, Jeff Gose and John Pierce at Santa Cruz lighthouse solar viewing session*

## Fall 2014 SJAA Swap Happiness

From Michael Packer

The 2014 SJAA Swap made many a stargazer happy! There is something beautiful about a swap meet that beckons back to the day of selling a sheep for wool coats. Shear Poetry, Ha! Well OK – anyway folks sold equipment that they no longer need, to folks who can't believe they're walking home with telescope, eyepiece, tripod or binocular.

On the club side of things – SJAA provided pizza and drink – and we collected a few donations. Stellar Cheers, Stellar Gear –Thanks go to all who helped to make this a special and successful event.

*Editor's Note: Rich Neuschaefer, Lee Hoglan, Sukhada Palav, Greg Claytor and Rob Jaworski volunteered to help run the swap meet. Along with a number of other SJAA members who showed up to help out. Photos below.*



# SJAA Member Observation Posts

Editor's note: Following below and on the next page are two good examples of different observing log styles which were posted on the SJAA observers Yahoo page. Mark McCarthy's log more empirical and Alexander Kramarov's log more of a streaming narrative. Both logs, rich in detail, made me feel I was right there with them. Well done.

## From Mark McCarthy

Noticing an internet forum post mentioning a partial transit of Io in front of Ganymede, I decided to attempt to view it even though it would be at 2am PST. I love observing transient events. It sounds corny but I have a feeling of connection with the universe seeing change in the heavens in my finite, human timescale.

I started observing at 2:05am (10:05 UT as in my notes). Seeing was 6/7 stable, and transparency was not too bad, average. A little dewy, temperature in the low 40s. After trying a range of magnifications I settled on 277x. Jupiter looked mushy but I could still see the moons clearly enough to use it to observe the transit.

To the right are my log notes with sketches. I added an exaggerated view to the right hand side to better show the progress of the transit. Most interesting was to see an apparent size difference between Io and Ganymede -- probably not noticeable unless they are very close to each other.

Scope: 12.5" f/7 newt; 8mm eyepiece @ 277x 0.4" TFOV. 95% full moon; Seeing 6/7 Transparency 3/5. Times in UT:

10:13: Transparency not too bad; Trapezium in Orion showing A-F stars at 170x. Ganymede and Io look like a nice double star of near equal brightness. Callisto is far out to the West. Europa not visible as it is in Jupiter's shadow

10:16: Split narrowing. Two distinct disks in contact, just separating when seeing stills. Ganymede is bigger than Io. Jupiter mushy at this magnification

10:18: Notched but not separating, figure "8"

10:21: Maximum overlap. Elongated egg shape. No split, no notching

10:24: Just touching, notched but not separating

10:26: Easy close split. Separation grows as I watch.

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### RASG Visual Observing Log

UT Date: 10/3/14 Time: 10:05-10:28 Activity: Ganymede/Io transit

Location: Home, Fremont CA

Conditions: Transparency 6/7 Seeing 6/7 Lying Visual Magnitude: 1.5-5.0 95% full moon

Object: Jupiter Cons: RA 5 11 5

Type: Major Size: Chart Ref: Dec 5 11 5

Instrument: 12.5" f/7 newt Eyepiece: 8mm Filter: --

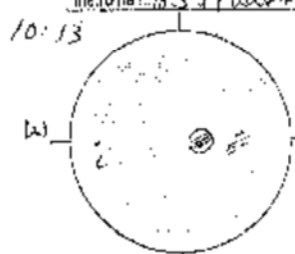
Notes: 277x 0.4" TFOV

Transparency not too bad

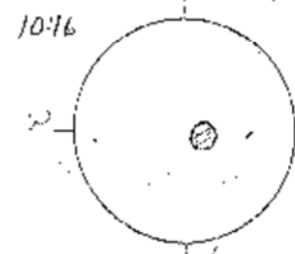
Trapezium A-F @ 170x

Ganymede and Io look like nice double star, near equal brightness. Callisto is far out to the West. Europa not visible as it is in Jupiter's shadow

10:13



10:16



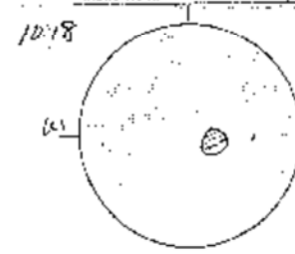
Split narrowing

Two distinct disks in contact, just separating when seeing stills

Ganymede is bigger than Io

Jupiter mushy at this mag.

10:18



Notched but not separating, figure "8"

Above image: Mark McCarthy's observation log

### From Alexander Kramarov

On November 22nd, due to a family commitment, I knew I would not be able to make it to Mendoza by 6:30pm, and I opted for the overflow lot over Hunting Hollow (higher is better). The windy road in the dark was daunting, but I scaled the mountain (breaking for a minute when a huge hog decided to walk on the road in front of me) and got to the lot by 7:30pm. Surprisingly, it was completely empty! I remember 3 months ago the lot was full on new moon weekend, I guess today's weather/temperatures did not help to get people out to observe.

This evening was a test of my winter observing abilities. Normally I am limited by 1) cold 2) hunger 3) laptop battery 4) dew on my finders, so since I had the whole summer to prepare, I was ready. I had 4 layers on my feet, 3 on my legs and head, 6 on my torso, and 2 layers of gloves. Some energy bars + half a gallon of hot tea helped as well, and since I had all of the lot to myself I was able to position the car so it would block the wind, keeping me even warmer than I wanted at times. The laptop was connected to a 110 AH battery, so it would probably last for >24 hours. The finders were covered by homemade dew shields made from a large pill box (custom cut to fit over the red dot finder) and an adjustable neoprene knee sleeve fitted over my 9x50. The 12" dob did not need dew protection.

At the beginning transparency was below average, I could only see dim stars up to about 30 degrees from zenith on north, west and south, everything else was washed out by the lights from SJ and Gilroy. The eastern horizon was darker and much better for observing so I decided to concentrate on Messier object search in the area between Taurus, Pisces, Cancer and Canis Major. In addition to covering the Messier objects, my goal was to identify every constellation I could, in that area and around it.

I started in and around Orion (my first look at M42 this winter). Next I jumped to M79 in Lepus; a very small globular, even in 9mm or 6mm (x254). I could not clearly resolve any stars. I then moved to Auriga and covered its three open clusters M36, M37, M38. M37 is amazing, as always, my favorite object. I then jumped to the magnificent M35 in Gemini's foot - a rich open cluster filling the FOV of a 32mm. Next, I took a look at M1 (crab nebula in Taurus), readily visible and quite large in a 32mm, but except from the elongated shape, no details were visible with or without nebula filter. I took a minute to revisit M45 (Pleiades), some nebulosity can be found around these very bright stars if you have seen the long exposure images and know what to look for. Two galaxies M74 in Pisces and M77 in Cetus can be easily found. For M74, if you continue in the direction of the bright stars in Aries until you hit the bright star in Pisces. M77 in Cetus can easily be found by following the line extending out of Taurus into the 3 bright stars of Cetus. M77 looks like a beautiful, bright galaxy, with some structure apparent. Both of these are now added to my list of imaging targets, surely a photo of these would reveal much more detail than I can see with my own eyes. At that moment I examined my maps to see if any planets

were up, and Uranus was in a good place to see the blue disk. Jupiter was still below the mountains so I moved to the lower constellations. Finding cancer is no easy task with all the dim stars, but it appears that the easiest way to find it is to look for the mag 3.1 M44, visible with the naked eye. That cluster is so large that it does not fit into my 32mm FOV. Another Messier object in Cancer is M67; quite a peculiar open cluster, looks like a short plant where you have a flower at the top, short stem, and a lot of roots below (yes, I realize that my dob provides a flipped image :).

Around this time Jupiter finally cleared the mountains, and to my surprise I only saw 2 moons. It turns out that Callisto was very close to Jupiter itself, and seeing so close to the horizon was so bad that I could not separate it from the huge disk of the planet itself. Io was actually behind Jupiter, and was scheduled to emerge from behind around 0:30am.

Looking at Stellarium, the whole area between Monoceros, Puppis and Canis Major is riddled with interesting objects. M41 in Canis Major is an appealing open cluster. M58 is not that interesting (formless open cluster filling half of 32mm FOV). M47 is a very bright open cluster, and is worth spending a few minutes to examine it. The nearby M46 consists of very dim stars, and a perceptive observer can find a planetary nebula near the edge of that cluster (nebula filter does help to see it). Definitely a target for long focal length imaging.

The heart cluster, M50 really looks like a heart (worth showing to your significant other). M78 is a small reflection nebula in Orion, although its images look interesting, visually I only spent about 15 seconds on it before I moved on since I could not see much detail. Last object of the night was M93 - an unremarkable open cluster in Canis Major.

When Leo's head came out behind Jupiter I decided to call it a night, my daughter's birthday is tomorrow (oh, I mean today, since it was after midnight), and I need to be fresh at the morning. I checked if there are any comets visible at that time, but the only one that I could find from my sources is C/2014 R1, in Virgo, and would be visible around 4am - too late for my observing plans. While I did not bag any special objects, the iota-Aurigids meteor shower provided quite frequent and welcome background "fireworks" for this evening.

On the way back down from the mountain a fox and 3 deer made themselves visible to my car's lights, so this was a really good sky and terrestrial observing evening. And all of the equipment I brought with me functioned as I expected, and knowing this, next time I can surely stay observing until sunrise, if I would ever decide on such a crazy endeavor.





## Kid Spot Jokes:

- ♦ **What do planets like to read?**  
(Comet books)
- ♦ **What do you call a story about a comet?**  
(A tail)

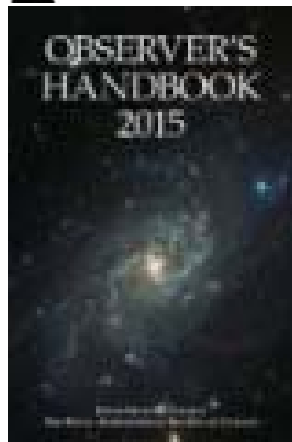
## Kid Spot Quiz:

1. **Comets get their name from the Greek phrase “aster kometes”**  
What does this mean?
2. **What is the hard head of the comet called?**

## Kid Spot Night Sky Challenge: December 2014

- ⇒ Mars—low in the southwestern sky
- ⇒ Jupiter in Leo
- ⇒ Saturn—Predawn
- ⇒ Geminid Meteor showers—Dec 13/14

<http://skyandtelescope.com/observing/ata glance>



## Constellations

### Pisces

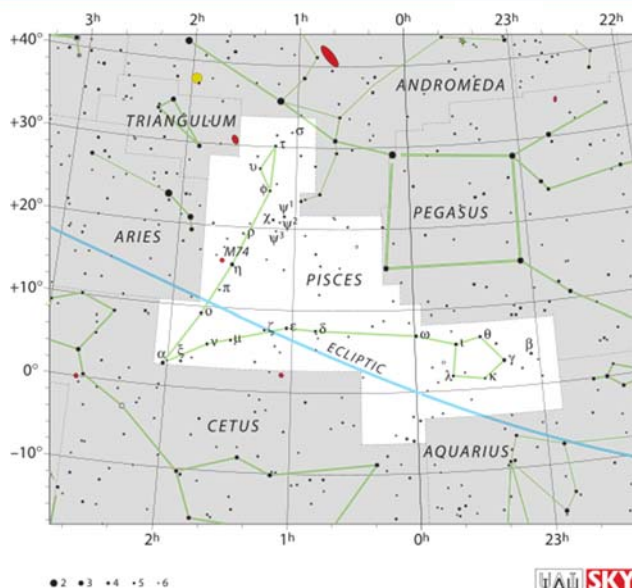
Pisces is a constellation of the zodiac. Its name is the Latin plural for fish, and its symbol is ♓. It lies between Aquarius to the west and Aries to the east.

Pisces is associated with Aphrodite and Eros, who escaped from the monster Typhon by leaping into the sea and transforming themselves into fish. In order not to lose each other, they tied themselves together with rope.

The ecliptic and the celestial equator intersect within this constellation and in Virgo. The Vernal equinox is currently located in Pisces, due south of  $\omega$  Psc, and, due to precession, slowly drifting below the western fish towards Aquarius.

#### Deep-sky objects

M74 is a loosely-wound (type Sc) spiral galaxy in Pisces, found at a distance of 30 million light years. It has many clusters of young stars and the associated nebulae, show-



ing extensive regions of star formation. It was discovered by Pierre Méchain, a French astronomer, in 1780. A type II -P supernova was discovered in the outer regions of M74 by Robert Evans in June 2003; the star that underwent the supernova was later identified as a red supergiant with a mass of 8 solar masses.

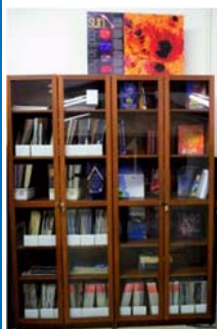
Credit:  
Wikipedia

#### Kid Spot Quiz Answers:

- 1) Aster means star and kometes means long-haired; comet = long-haired star
- 2) The nucleus



## SJAA Library!



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers!

Please check out our

wish list on the SJAA webpage:

<http://www.sjaa.net/sjaa-library/>

## Quick STARt Program

The Quick STARt Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

## SJAA at Lick Observatory

Bill O'Neil contacted Lick. There is potential for a private star party, max 30 people. Bill will follow up with Lick on possible future events.

## Fix It Program

Ed Wong plans to step down as program chair over the next few months. Vini Carter agreed to step in. Ed will continue to support Fix It during the transition.

## Advanced Loaner Program

Manoj Koushik has agreed to take over the Advanced Loaner Program from Dave Ittner. Dave will assist during the transition.

## From the Board of Directors

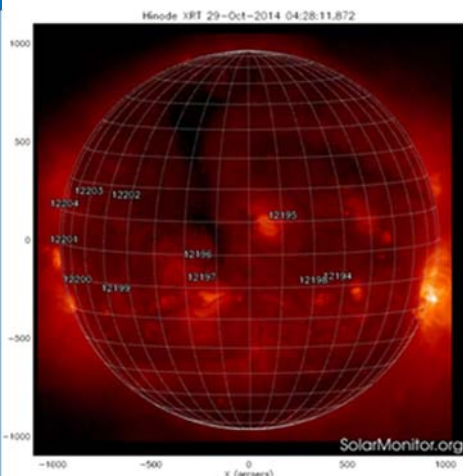
### Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

Weather doesn't slow this program down, so even if it's windy, cold and rainy outside, Fix It day goes on. Big thanks to go Ed Wong, Phil Chambers, and Dave Ittner for being the gear experts who faithfully make themselves available at Fix It day and a big welcome to Vini Carter who has stepped up to take over as Fix It program chair from Ed Wong.

### Solar Observing

Solar observing sessions, headed up by Michael Packer, are held the 1st Sunday of every Month from 2:00-4:00 PM at Houge Park weather permitting. See article and photos on page 5.



### Announcements

Annual Board elections are coming at the Membership Meeting this February 2015. The following Board members are up for re-election: Lee Hoglan, Michael Packer, Teruo Utsumi, Rich Neuschaefer and Ed Wong. If anyone is interested in running, please contact any board member for more information.

### Board Meeting Excerpts November 08, 2014

#### In attendance

Rob Jaworski, Lee Hoglan, Dave Ittner, Rich Neuschaefer, Bill O'Neil, Teruo Utsumi

#### Fulfilling RASC Pre-Orders

Teruo Utsumi ordered 35 each of RASC handbooks and calendars based on the online pre-order form. Delivery is expected soon.

#### New Ephemeris Schedule

Dec 14 is last monthly edition. Quarterly schedule will start March 2015. The issues will be published just before the last month of each quarter.

#### Winter Deep Sky Scavenger Hunt

Jan 16/17 Feb 13/14 are the planned dates for the DSSH. Ed Wong asked to start advertising in December. There will be all new objects to find! As in prior DSSH events, Fridays will have coaches and green laser pointers allowed.

#### Mendoza Ranch Update

The permit is renewed with 24 dates for 2015.

### SJAA Contacts

|                        |                  |
|------------------------|------------------|
| President:             | Rob Jaworski     |
| Vice President:        | Lee Hoglan       |
| Treasurer:             | Michael Packer   |
| Secretary:             | Teruo Utsumi     |
| Director:              | Greg Claytor     |
| Director:              | Dave Ittner      |
| Director:              | Ed Wong          |
| Director:              | Rich Neuschaefer |
| Director:              | Bill O'Neil      |
| Beginner Class:        | pending          |
| Fix-it Program:        | Vini Carter      |
| Imaging SIG:           | Mark Striebeck   |
| Library:               | Sukhada Palav    |
| Loaner Program:        | Manoj Koushik    |
| Ephemeris Newsletter - |                  |
| Editor:                | Sandy Mohan      |
| Prod. Editor:          | Tom Piller       |
| Publicity:             | Rob Jaworski     |
| Questions:             | Lee Hoglan       |
| Quick STARt            | Dave Ittner      |
| Solar:                 | Michael Packer   |
| School Events:         | Jim Van Nuland   |
| Speakers:              | Teruo Utsumi     |

E-mails:

<http://www.sjaa.net/contact>

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by the 20th of the month of February, May, August and November.

San Jose Astronomical Association  
P.O. Box 28243  
San Jose, CA 95159-8243  
<http://www.sjaa.net/contact>

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## San Jose Astronomical Association Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

☐ **New** ☐ **Renewal** (Name only if no corrections)

### Membership Type:

- ☐ Regular — \$20  
☐ Regular with Sky & Telescope — \$53  
☐ Junior (under 18) — \$10  
☐ Junior with Sky & Telescope — \$43

Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2-3 months lead time.)

☐ **I prefer to get the Ephemeris newsletter in print form (Add \$10 to the dues listed on the left). The newsletter is always available online at:**

<http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to  
[sjaamemberships@gmail.com](mailto:sjaamemberships@gmail.com)

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